Strategic Data Source Integration for "Fessi": A Knowledge Graph Architecture for Waste Wood and Furniture Management in German Higher Education

1. Executive Summary and System Architecture Overview

The development of "Fessi," a specialized knowledge graph and conversational agent for waste management within a German university context, represents a significant technical and ecological undertaking. The domain of waste wood (*Altholz*) and furniture disposal is not merely a logistical challenge but a complex intersection of material science, historical industrial chemistry, strict legislative frameworks, and circular economy principles. Unlike standard municipal waste streams where visual identification is often sufficient, the categorization of waste wood requires a forensic approach to data. A seemingly innocuous wooden beam may harbor persistent organic pollutants from the 1970s, or a modern office desk may represent a composite material puzzle requiring disassembly protocols to maximize recycling potential.

This report provides an exhaustive analysis of the data landscape required to power Fessi's reasoning engine. It moves beyond superficial waste lists to establish a deep semantic infrastructure that connects high-level regulatory statutes—specifically the German Waste Wood Ordinance (*Altholzverordnung - AltholzV*)—with granular, instance-level data regarding material composition, historical preservative usage, and regional disposal workflows. The objective is to construct a knowledge graph that does not simply parrot static rules but actively reasons about the probable contamination status of an object based on its provenance, visual attributes, and physical composition.

The necessity for such a system is underscored by the regulatory rigidity of the German waste sector. The misclassification of hazardous waste wood (Category A IV) as recyclable material (Category A I or A II) poses severe environmental risks, including the release of dioxins and heavy metals during reprocessing. Conversely, the incineration of high-quality, untreated wood represents a failure of the circular economy and a loss of valuable secondary raw materials. Fessi must navigate this dichotomy by integrating diverse data sources: from the technical XML schemas of the Ökobaudat environmental database to the unstructured,

legacy-laden PDFs of university waste guidelines.

This document serves as the foundational architectural blueprint for Fessi. It delineates the specific data points, API structures, and ontological relationships necessary to build a system capable of discerning the subtle difference between a chemically treated railway sleeper from the 1960s and a modern, heat-treated Euro-pallet. By synthesizing these disparate information streams into a cohesive knowledge graph, Fessi will empower university stakeholders—from facility managers to students—to make disposal decisions that are legally compliant, ecologically sound, and operationally efficient.

2. The Regulatory Ontology: Deep Dive into the Altholzverordnung (AltholzV)

The logic governing Fessi must be rigorously anchored in the legal definitions that dictate the flow of waste wood in Germany. The *Altholzverordnung* (AltholzV) is the primary legislative instrument, and its categorization system (A I to A IV, plus PCB-wood) serves as the top-level class hierarchy for the knowledge graph. Understanding the nuanced definitions within this ordinance is prerequisite for designing the decision trees Fessi will utilize.

2.1 The Categorization Hierarchy and its Data Implications

The AltholzV establishes a hierarchy based on the degree of chemical contamination. This is not a sliding scale but a set of distinct buckets with specific disposal mandates. Fessi's primary function is to map a user's natural language description of an object to one of these buckets.

2.1.1 Category A I: Natural and Mechanically Processed Wood

This category represents the "gold standard" for material recycling. It encompasses waste wood in its natural state or wood that has only been mechanically processed—cut, planed, or sanded—without the application of any chemical substances.

- **Ontological Definition:** In the Fessi ontology, an A I classification requires the negation of all chemical treatments. It is defined by what it lacks: no paint, no lacquer, no glaze, and absolutely no biocidal impregnation.
- **Typical Entities:** The data suggests that Euro-pallets (*Europaletten*) made of solid wood are the archetypal A I item, alongside transport crates made of solid timber, fruit crates, and cable drums manufactured after 1989.[1]
- **Disposal Pathway:** This material is highly sought after for the production of wood chips for the particleboard industry. Energetic recovery is permitted, but material recovery is prioritized.
- **Data Signal for Fessi:** The chatbot must explicitly query for "treatments." If a user identifies an object as "bright," "raw," or "untreated," and the object type aligns with known A I profiles (e.g., "pallet"), the confidence score for A I should be high. However,

Fessi must account for subtle contamination; for instance, a pallet used to transport chemicals might be contaminated despite being structurally A I.

2.1.2 Category A II: Glued, Painted, or Coated Wood (Non-Hazardous)

Category A II accounts for a massive volume of municipal and construction waste. It includes wood that has been treated with glues, paints, lacquers, or coatings, provided these substances do not contain halogenated organic compounds and the wood has not been treated with wood preservatives.

- **Ontological Definition:** This class admits chemical treatments but restricts them to non-hazardous binders and finishes. It effectively covers "interior wood."
- **Typical Entities:** Interior door leaves (*Türblätter*), floorboards (*Dielen*), engineered wood products like particleboard (*Spanplatten*), plywood, and OSB, as well as skirting boards and ceiling panels used in interior construction.[1]
- **Disposal Pathway:** These materials can be recycled into particleboard, subject to strict cleaning and sorting processes to remove metals and plastics, or used for energy generation in biomass plants equipped with appropriate filtration.
- **Data Signal for Fessi:** The keyword "Interior" (*Innenbereich*) is the strongest heuristic for A II. Fessi needs to link terms like "varnished shelf" or "laminate flooring" to this category. The distinction from A III rests largely on the chemical nature of the coating (e.g., PVC vs. Acrylic).

2.1.3 Category A III: Wood with Halogenated Coatings

This category is technically distinct but often managed alongside A II in modern incineration contexts. It covers waste wood with coatings that contain halogenated organic compounds (Chlorine, Bromine, Fluorine) but no wood preservatives.

- **Ontological Definition:** The defining attribute is the presence of halogens in the surface layer. This typically points to Polyvinyl Chloride (PVC) foils.
- **Typical Entities:** Furniture with PVC coatings, older kitchen furniture (where PVC foils were common for durability), and wood from acoustic panels.[1]
- **Disposal Pathway:** Material recycling is generally prohibited due to the chlorine content, which can cause corrosion in processing equipment and dioxin formation. Energetic recovery is restricted to plants with advanced flue gas cleaning systems capable of scrubbing acid gases.
- **Data Signal for Fessi:** Fessi must identify "plastic-coated" items. A "kitchen cabinet" with a "glossy plastic front" should trigger an A III probability. The system should flag the presence of "PVC" or "Vinyl" as a deterministic marker for A III.

2.1.4 Category A IV: Hazardous Waste Wood

Category A IV represents the highest risk tier and is the most critical for Fessi to correctly identify to prevent environmental contamination. This category includes wood treated with wood preservatives (fungicides, insecticides) and generally encompasses structural wood

used in exterior applications.

- **Ontological Definition:** The presence of biocidal agents is the defining feature. Because these agents are designed to be persistent and toxic, the wood remains hazardous essentially forever.
- **Typical Entities:** Railway sleepers (*Bahnschwellen*), telegraph poles, hop poles, vineyard stakes, garden fences, terrace decking, window frames (almost universally A IV due to historical preservation standards), and external doors.[1]
- **Disposal Pathway:** Strict incineration in hazardous waste incinerators or specialized biomass power plants compliant with the strict emission limits of the 17th BImSchV (Federal Immission Control Ordinance). Material recycling is strictly prohibited.
- **Data Signal for Fessi:** The concept of "Outdoor Wood" (*Außenbereichsholz*) is the primary trigger. Any wood exposed to the elements is presumed to be treated. Historical markers (discussed in Section 3) are vital here.

2.1.5 PCB-Contaminated Wood

A special sub-category exists for wood treated with agents containing Polychlorinated Biphenyls (PCB).

- **Ontological Definition:** Wood containing >50 mg/kg of PCBs.
- **Typical Entities:** Acoustic panels, insulation plates, and certain window sealants from the 1960s and 1970s.
- **Data Signal for Fessi:** This requires a temporal query. If the building waste originates from the 1960s or 1970s, Fessi must flag the potential for PCB contamination and advise special testing.[3]

2.2 Operationalizing the Waste List Ordinance (AVV)

The *Abfallverzeichnis-Verordnung* (AVV) provides the numerical codes required for all official documentation, such as transport manifests (*Begleitscheine*) and disposal records. Fessi must function as a translation layer, converting the user's colloquial description of an object into the precise AVV code required by the university's facility management.

The following table synthesizes the mapping between physical objects, their usage context, and the corresponding AVV codes, highlighting the conditional logic Fessi must implement.

Table 1: Semantic Mapping of Wood Waste to AVV Codes

Physical Object	Condition / Context	AVV Code	Category	Hazardous ?	Rationale / Data Source

Sawdust / Shavings	Pure, untreated wood	**03 01 05**	A I	No	Industrial residue, clean [7]
Sawdust / Shavings	Containing hazardous substances	**03 01 04***	A IV	**YES**	Contaminated processing residue [7]
Pallet (Euro/Ind)	Solid wood, untreated	**15 01 03**	A I	No	Packaging waste [1]
Cable Drum	Manufacturing Date > 1989	**15 01 03**	A I	No	Modern production standards [8]
Cable Drum	Manufacturing Date < 1989	**17 02 04***	A IV	**YES**	Presumed treated (e.g., with arsenicals) [8]
Construction Wood	Untreated / Painted (Interior)	**17 02 01**	A I / A II	No	Standard renovation waste [3]
Construction Wood	Impregnated / Exterior Use	**17 02 04***	A IV	**YES**	Biocidal treatment presumed [3]
Window Frame	All Types (Standard presumption)	**17 02 04***	A IV	**YES**	Preservation + Glazing compounds [5]
Railway Sleeper	Tar Oil (Creosote)	**17 02 04***	A IV	**YES**	High PAH content [5]
Insulation Board	PCB-containing	**17 06 03***	PCB	**YES**	Hazardous insulation

					material [7]
Furniture (General)	Mixed bulk waste	**20 03 07**	Sperrmüll	No	Aggregate category for collection [7]
Furniture (Wood)	Separately collected	**20 01 38**	A I - A III	No	Targeted for recycling [3]

Note: Codes marked with an asterisk (*) denote hazardous waste requiring specific handling protocols.

This table illustrates the complexity Fessi must navigate. A "Cable Drum" is not a single entity in the graph; it is a class with a temporal property ("year of manufacture") that determines its hazard status. Similarly, "Construction Wood" splits into two distinct legal realities based on its installation location (Indoor vs. Outdoor).

3. Historical Forensic Analysis: The Chemistry of Legacy Wood

A significant challenge for Fessi is that a large proportion of waste wood entering the disposal stream today originates from buildings and structures erected decades ago. This "legacy wood" carries the chemical fingerprint of the era in which it was produced. In Germany, this is further complicated by the division between the Federal Republic of Germany (BRD) and the German Democratic Republic (DDR), which had distinct chemical industries, regulatory standards, and preservative formulations. Fessi must act as a "historical forensic analyst," using the provenance of the wood to infer potential contamination.

3.1 The East/West Chemical Divide

The divergence in wood preservation practices between East and West Germany until reunification in 1990 creates a distinct geographical and temporal logic for waste classification.

3.1.1 The DDR Legacy: Hylotox and DDT

In the former East Germany (DDR), the chemical industry was heavily reliant on specific organochlorine compounds that were phased out earlier or used differently in the West.

- **Hylotox 59:** This is the most critical historical preservative for Fessi to recognize in East German contexts. It was an oily wood preservative containing **DDT** (Dichlorodiphenyltrichloroethane) and **Lindane** (Gamma-Hexachlorocyclohexane).

    - **Usage Profile:** It was widely used for roof trusses (*Dachstühle*), ceiling beams, and even indoor furniture to combat wood-boring insects.
    - **Timeline:** Its usage was prevalent until 1989/1990.
    - **Identification:** Wood treated with Hylotox 59 often appears oily or dark, but the active ingredients are invisible. The primary signal is context: "East German building" + "Roof timber" + "Pre-1990."
    - **Health Implications:** DDT is a persistent organic pollutant (POP). It accumulates in adipose tissue, is carcinogenic, and acts as an endocrine disruptor. Its presence mandates disposal as hazardous waste (A IV) to ensure destruction in high-temperature incineration.[9]
- **Salt-Based Agents (Ull-Salts):** The DDR also utilized salt mixtures known as "Ull-Salts" (*U-Salze*), such as **Dohnalit**. These typically contained Fluorine and Chromium compounds.
    - **Visual Cue:** These treatments often imparted a pink or green hue to the wood, a vital visual feature for Fessi to query.[9]

3.1.2 The BRD Legacy: Xylamon and PCP

In West Germany, the dominant wood preservatives for decades were based on **Pentachlorophenol (PCP)**.

- **Xylamon / Xyladecor:** These trade names are synonymous with West German wood protection. Until the late 1970s and early 1980s, formulations frequently contained PCP and Lindane.
- **Timeline:** The widespread use of PCP continued until the "PCP Ban Ordinance" (*PCP-Verbotsverordnung*) was enacted in 1989. This date serves as a hard cutoff in Fessi's logic: Wood installed in the BRD *after* 1990 is significantly less likely to contain PCP.
- **Visual Cue:** High concentrations of PCP and Lindane can sometimes "bloom" on the surface of old wood, forming "Raureif" (hoarfrost-like) crystalline deposits. Fessi should prompt users to look for "white crystals" on very old, dark wood.[5]
- **Health Implications:** PCP is highly toxic and, when burned at low temperatures (e.g., in a domestic fireplace), generates dioxins and furans—some of the most toxic compounds known. This underscores the absolute necessity of routing such wood to professional A IV disposal facilities.

3.2 Detailed Preservative Database for Fessi

To support this forensic reasoning, Fessi's backend must include a database of historical preservatives. The following table details specific agents, their active ingredients, and their associated hazard profiles.

Table 2: Historical Wood Preservatives and Active Ingredients for Knowledge Graph Nodes

Trade Name	Region	Active Ingredients	Typical Application	Usage Period	Hazard Class	Data Sources
Hylotox 59	DDR	DDT, Lindane	Roof trusses, Furniture, Indoors	Until ~1989	A IV	[9]
Hylotox IP	DDR	Pentachlorophenol (PCP)	Exterior structural wood	1960s-1980s	A IV	[9]
Xylamon	BRD	PCP, Lindane	Structural timber, Furniture	Until ~1978/84	A IV	[5]
Xyladecor	BRD	PCP, Lindane	Exterior cladding, Fences	Until ~1978/84	A IV	[5]
Basilit	Both	Fluorine, Chromium, Arsenic	Heavy construction timber	1920s-1980s	A IV	[9]
Carbolineum	Both	Tar oils (PAHs)	Sleepers, Poles, Sheds	Until ~1991	A IV	[5]
Donarolit	BRD	Dinitrophenol	Construction timber	Mid-20th Century	A IV	[5]
Kombinal TO	DDR	TBTO (Organotin), Thiram	Construction, Windows	1970s-1980s	A IV	[9]

Dohnalit	DDR	Fluorine, Chromium salts	Roofs, structural wood	1960s-1990	A IV	[9]

3.3 Tar Oils and PAHs

A distinct category of historical preservatives involves **Tar Oils** (Carbolineum), which are distillates of coal tar.

- **Chemical Profile:** These contain high concentrations of **Polycyclic Aromatic Hydrocarbons (PAHs)**, including the carcinogen Benzo[a]pyrene.
- **Sensory Identification:** Tar oil treated wood is unique in that it can often be identified by smell. It emits a strong, characteristic "tarry" or "railway track" odor, which can persist for decades. Visually, the wood is dark brown to black and may feel sticky or oily to the touch, especially in warm weather.
- **Fessi Application:** Fessi should explicitly ask about odor: "Does the wood smell like tar or old railway tracks?" A positive response is a near-definitive indicator for A IV classification.[5]

4. Material Science of Waste Wood: Modern Identification and Analysis

While historical wood presents a chemical hazard, modern waste wood presents a challenge of material complexity. Fessi must navigate the identification of composites, visual markers of pressure treatments, and the integration of new material data.

4.1 Visual Identification Framework

Since Fessi relies on user input, it must translate complex chemical realities into visual questions. The knowledge graph must include an "Image Feature" or "Visual Cue" node type linked to specific contamination probabilities.

4.1.1 Color-Based Indicators

- **Green Tint:** This is the hallmark of modern pressure-treated wood. It indicates the presence of **Copper-based preservatives** (e.g., CCA - Chromated Copper Arsenate, CCB - Copper Chrome Boron, or ACQ - Alkaline Copper Quaternary).
  - *Interpretation:* While modern variants (ACQ) are less toxic than historical arsenic-containing ones, for the purpose of waste sorting, **Green = Treated = A IV**. Fessi must rigorously enforce this rule.
  - *Context:* Commonly found in garden fences, playground equipment, carports, and terrace decking.[12]
- **Blue Tint:** A blue coloration can indicate two distinct things.
  - *Preservative:* Boron-based treatments are sometimes dyed blue for identification

(e.g., "Advance Guard"). Boron is generally less toxic to mammals but is considered a contaminant in recycling streams if not washed out.
    - *Fungal Stain:* "Blue Stain Fungi" can naturally discolor wood. Fessi needs to help the user distinguish between a natural fungal stain (often irregular, grayish-blue) and an artificial dye (uniform, bright blue).
- **Dark Brown / Black:** As noted, this often indicates **Tar Oil (Creosote)** or pigmented stains. If the wood is oily, it is likely Creosote (A IV). If it is dry and merely painted, it may be a decorative glaze (A II).[13]

4.1.2 Structural and Surface Cues

- **Injection Incisions:** Industrial pressure treatment (*Kesseldruckimprägnierung*) often involves mechanically incising the wood to allow deeper penetration of the preservative.
    - *Fessi Signal:* "Do you see rows of small, regular cuts or holes on the surface?" -> Positive response indicates A IV.[12]
- **Crystalline Deposits:** The "Raureif" effect mentioned earlier is a key indicator for aged PCP/Lindane treated wood.
- **Manufacturer Stamps:**
    - **"EUR" / "EPAL":** Found on pallet blocks. Indicates quality-controlled production, generally A I (unless stained/dirty).
    - **"MB":** Methyl Bromide. Found on older packing wood. Indicates fumigation with a toxic gas.
    - **"HT":** Heat Treated. Indicates physical sterilization without chemicals. This is a positive indicator for A I classification.[12]

4.2 Chemical Testing Data Integration

While Fessi interacts with users who likely lack laboratory equipment, the system architecture should accommodate data from simple field tests, should the university equip its waste centers with them.

- **PAN Stain Test:** This uses 1-(2-pyridylazo)-2-naphthol indicator.
    - *Reaction:* Turns **Magenta/Red** in the presence of Copper.
    - *Utility:* Confirms the presence of CCA, CCB, or Copper organics in "greenish" wood, solidifying the A IV classification.[14]
- **Beilstein Test:** A classic test for halogens (PVC). A copper wire is heated and touched to the plastic coating, then returned to the flame. A green flame indicates Chlorine. Fessi could contain this as "Expert Level" advice for facility managers identifying A III plastics.

4.3 Composite Materials and "Wood-Plastic" Hybrids

The line between "wood" and "plastic" is blurring, creating new waste streams that fit poorly into the classic AltholzV categories.

- **WPC (Wood Plastic Composites):** Used increasingly for terrace decking.

    - *Composition:* A mix of wood fiber and thermoplastics (PE, PP, PVC).
    - *Disposal:* WPC is **not** pure Altholz. Depending on the plastic content, it may be rejected by wood recyclers and require disposal as residual waste or bulky waste for thermal treatment. Fessi must explicitly identify "plastic-like wood" to prevent contamination of the wood recycling stream.[3]
- **Glues and Resins:** Formaldehyde emissions are a key concern in particleboards. While modern boards adhere to the E1 emission standard, university waste streams often contain "legacy furniture" from the pre-regulation era. These older boards have high formaldehyde potential and belong in A II for energy recovery in certified plants, rather than material recycling.[17]

5. Furniture and Bulky Waste Stream Analysis: The "Sperrmüll" Challenge

For a university, the "Sperrmüll" (bulky waste) stream is a dominant operational reality. Dormitory clearances, office renovations, and lecture hall updates generate massive volumes of furniture. Fessi must treat "furniture" not as a monolith but as a complex assembly requiring deconstruction guidance.

5.1 Life Cycle Assessment (LCA) and Material Composition

Data from LCA studies provides the "recipe" for modern office furniture, allowing Fessi to predict the material breakdown of an item.

- **Office Desks:**
    - *Composition:* Typically dominated by a particleboard work surface (A II) and steel legs. The core is often ~90% wood chips and 10% resin binder.
    - *Energy Implication:* The production of a workspace consumes significant energy (~1500-2400 MJ). Recycling the steel legs saves substantial energy compared to virgin steel production.[17]
    - *Fessi Action:* Fessi should advise: "Unscrew the metal legs. Place legs in the Scrap Metal container and the wooden top in the Wood A II container." This simple sorting instruction, derived from compositional data, significantly increases the recycling rate.
- **Office Chairs:**
    - *Complexity:* These are highly heterogeneous, containing steel, aluminum, polypropylene, polyurethane foam, and fabric.
    - *Recyclability:* Due to the tight integration of materials (e.g., foam molded onto plastic), manual separation is difficult.
    - *Classification:* Often destined for the "Sperrmüll" container for mechanical shredding and magnetic separation, rather than pure wood recycling.[19]

5.2 Sperrmüll Composition Statistics

Understanding the aggregate composition of bulky waste helps in planning container capacity and estimating disposal costs.

- **Wood Fraction:** Studies indicate that wood constitutes between **17% and 30%** of municipal bulky waste by weight.[21]
- **Furniture Dominance:** Within this wood fraction, furniture is the dominant component, often accounting for over **50% of the volume**.[23]
- **Recycling Potential:** Analysis suggests that up to **82%** of Sperrmüll weight could be recycled if rigorously separated into Wood, Metal, and Plastic fractions. However, current practices often lead to incineration due to poor sorting at the source.[21]
- **Trend Analysis:** The "Fast Furniture" trend is altering the waste composition. The proportion of solid wood (A I) in furniture waste is decreasing, replaced by lower-grade composites (A II) and hollow-core structures (honeycomb paper fill), which have lower calorific value and different recycling properties.[25]

6. Technical Architecture: Data Source Integration for the Knowledge Graph

To operationalize this domain knowledge, Fessi must ingest data from specific, high-quality sources. This section details the technical "supply chain" for Fessi's data.

6.1 The Ökobaudat API: Environmental Product Data

The **Ökobaudat** platform, maintained by the German government, is the premier source for Life Cycle Assessment (LCA) data and Environmental Product Declarations (EPDs). It provides a machine-readable, standardized description of construction materials.

- **Data Standard:** The database utilizes the **ILCD+EPD** format, an XML-based schema compliant with EN 15804. This allows for precise, structured queries about material properties.
- **API Architecture:**
  - *Endpoint:* https://www.oekobaudat.de/api/datasets
  - *Access Method:* REST API.
  - *Data Retrieval:* Fessi can query datasets by UUID or search terms (e.g., "Spanplatte", "Fensterrahmen").
- **Integration Strategy:**
  1. **Ingestion:** Use a Python interface (using requests and XML parsing libraries) to fetch relevant datasets.
  2. **Mapping:** Extract fields such as flow properties (density, mass), material composition (percentage of binders vs. wood), and embedded waste codes.
  3. **Application:** When a user asks about the "sustainability" or "carbon footprint" of

disposing of a specific wood type, Fessi can pull the Global Warming Potential (GWP) data from the corresponding EPD.[26]

6.2 Wikidata and the Semantic Web

Wikidata offers a global, multilingual ontology that can serve as the backbone for Fessi's entity resolution.

- **Entity Linking:** Fessi should link its internal nodes to Wikidata Q-IDs.
  - *Reclaimed Lumber:* **Q7302557**.[30]
  - *Preservatives:* **Q163648** (DDT), **Q207050** (Lindane), **Q423985** (Pentachlorophenol).
- **SPARQL Queries:** Fessi can use SPARQL to infer relationships or fetch translations.
  - *Example Query:* SELECT?hazard WHERE { wd:Q423985 wdt:P279?hazard } (Find the hazard class of Pentachlorophenol).
- **Utility:** This connects Fessi to the broader semantic web, allowing it to "know" that Lindane is an insecticide without needing a hardcoded local definition.

6.3 Localized Rules: University Abfall-ABCs

Generic regulations must be overlaid with local rules. University waste guidelines (e.g., University of Freiburg or Giessen) often exist as semi-structured PDFs or HTML lists.

- **Data Extraction:** These documents, often titled "Abfall-ABC" (Waste A-Z), list items mapping to specific disposal paths.
  - *Example Rule:* "Holz (Bauholz)" -> "Container A I-III" (if clean) or "Sonderentsorgung" (if treated).
  - *Contacts:* They provide crucial operational data, such as "Contact Department 4.5 for furniture removal".[31]
- **Ingestion Logic:** A scraper/parser is required to convert these "Item -> Disposal Path -> Contact" tuples into Knowledge Graph triples.

7. Knowledge Graph Ontology Design

The effectiveness of Fessi relies on a robust ontology that models the relationships between materials, treatments, regulations, and disposal actions.

7.1 Core Classes and Hierarchy

- **Class: Material**
  - *Subclasses:* SolidWood, CompositeWood, Plastic, Metal, WPC.
- **Class: ChemicalAgent**
  - *Subclasses:* Biocide (DDT, Lindane, PCP), Binder (Urea-Formaldehyde), Coating (PVC, Lacquer).
- **Class: WasteCategory**
  - *Instances:* Category_AI, Category_AII, Category_AIII, Category_AIV, PCB_Wood,

Sperrmüll.
- **Class: Treatment**
  - *Instances:* PressureImpregnation, Painting, Lamination, HeatTreatment.
- **Class: VisualFeature**
  - *Instances:* GreenTint, OilySurface, InjectionHoles, ManufacturerStamp.

7.2 Semantic Relations (Object Properties)

- hasMaterialComposition: Links an Object to a Material.
- wasTreatedWith: Links an Object to a ChemicalAgent or Treatment.
- hasVisualCue: Links an Object to a VisualFeature.
- originatesFrom: Links an Object to a Region (DDR/BRD) or Era (Pre-1990/Post-1990).
- legallyClassifiedAs: Links an Object to a WasteCategory.
- requiresDisposalAction: Links a WasteCategory to a defined Action (e.g., "Incinerate in hazardous waste plant").

7.3 Inference Logic (SWRL Rules)

The power of the graph lies in its ability to infer classification from partial data using Semantic Web Rule Language (SWRL).

- The "Outdoor" Rule:
  WoodObject(?w) ^ usedInContext(?w, Outdoor) -> legallyClassifiedAs(?w, Category_AIV)
  - *Reasoning:* Outdoor use implies preservative treatment.
- The "Hylotox" Rule (Historical Inference):
  WoodObject(?w) ^ originRegion(?w, DDR) ^ manufactureYear(?w,?y) ^ swrlb:lessThan(?y, 1990) ^ isPartOf(?w, RoofStructure) -> hasProbabilityOfContamination(?w, High_DDT)
- The "PVC" Rule:
  Furniture(?f) ^ hasCoating(?f, PVC) -> legallyClassifiedAs(?f, Category_AIII)

8. Visual and Sensory Identification Framework for the Chatbot

Fessi's user interface will be a chatbot. The backend data must therefore be structured to support a decision-tree dialogue based on sensory inputs.

8.1 The "Senses" Module

The system should prompt users to use their senses to identify hazards.

- **Smell:**
  - *Prompt:* "Does the wood have a strong, chemical, tar-like, or mothball smell?"
  - *Logic:* Yes -> **Carbolineum/Tar Oil** -> **A IV**.
- **Touch:**
  - *Prompt:* "Is the surface sticky or oily, especially if it's warm?"

    - *Logic:* Yes -> **Tar Oil** -> **A IV**.
    - *Prompt:* "Rub the surface. Is it rough/fibrous (Sawn) or smooth (Planed/Sanded)?"
    - *Logic:* Rough/Sawn often indicates construction timber; Smooth often indicates furniture/interior.
- **Sight:**
    - *Prompt:* "Do you see a greenish or olive tint?"
    - *Logic:* Yes -> **Copper Salts** -> **A IV**.
    - *Prompt:* "Do you see white, frost-like crystals on the surface?"
    - *Logic:* Yes -> **PCP/Lindane Blooming** -> **A IV** (High Risk).

8.2 The "Deconstruction" Module

For furniture, the chatbot must guide disassembly to optimize the AVV assignment.

- **Scenario:** A student wants to dispose of an old wardrobe.
- **Fessi Logic:**
    1. "Is the wardrobe made of solid wood or chipboard?" (Visual check of edges/grain).
    2. "Does it have a plastic coating?" (Gloss/Texture check).
    3. **Instruction:** "Remove the metal hinges and handles (Scrap Metal). If the wood has a plastic foil, it goes to A III (or Sperrmüll). If it's painted chipboard, it goes to A II."

9. Conclusion

The "Fessi" project represents a sophisticated application of knowledge engineering to the domain of environmental management. By moving beyond static lists and embracing a dynamic, graph-based model, Fessi can navigate the complex temporal and chemical landscape of German waste wood.

The integration of the **Altholzverordnung** provides the rigid legal skeleton, while the "historical forensics" of **DDR/BRD chemical history** provide the necessary flesh to identify legacy hazards like Hylotox and Xylamon. The utilization of the **Ökobaudat API** anchors the system in standardized, machine-readable environmental data, and the incorporation of **local university guidelines** ensures operational relevance.

Ultimately, Fessi does more than just sort trash; it digitizes the expertise of a waste manager. It translates the invisible threat of a 40-year-old wood preservative into a clear, safe, and compliant instruction for a student, thereby closing the loop on the circular economy and protecting the university community from environmental risks.

10. Appendix: Reference Tables and Decision Matrices

10.1 Visual Identification Matrix for Wood Waste

Visual Feature	Probable	Likely Category	Recommended

	Treatment		Action
Natural, bright wood	None	**A I**	Recycle (Material Recovery)
Painted (White/Color)	Lacquer / Acrylic	**A II**	Recycle (Energy Recovery)
Green / Olive Tint	Copper Salts (CCA, ACQ)	**A IV**	Hazardous Disposal (Incineration)
Black / Dark Brown / Oily	Tar Oil (Creosote)	**A IV**	Hazardous Disposal (Incineration)
Plastic Film (Wood grain)	PVC / Melamine	**A III / A II**	Energy Recovery
Hoarfrost Crystals	PCP / Lindane (Legacy)	**A IV**	**DANGER** - Avoid Contact
Incision Marks	Pressure Impregnation	**A IV**	Hazardous Disposal

10.2 Key Data Sources Summary

Source Name	Type	Key Data Provided	Relevance
Altholzverordnung	Regulation	Categories A I - A IV, Definitions	**High** (Core Logic)
AVV (Annex III)	Regulation	Waste Codes (17 02 01, etc.)	**High** (Compliance)
Ökobaudat	Database/API	EPDs, Material Composition, LCA Data	**Med** (Material Props)

University Abfall-ABC	PDF/Web	Local Disposal Rules, Contacts	**High** (User Context)
Historical Literature	Text	Hylotox, Xylamon, DDR/BRD Preservatives	**High** (Hazard ID)
Wikidata	Ontology	Semantic Linking, Multilingual Definitions	**Med** (Interoperability)

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